Suspensions of particles

Colloidal interactions, stabilisation and the Krieger–Dougherty model

Dr. Teodor Burghelea, HDR

CNRS Director of Research

Laboratoire de Thermique et Énergie de Nantes (LTeN), UMR 6607
CNRS – Nantes Université, Polytech Nantes

Teodor.Burghelea@univ-nantes.fr

Flows of Complex Fluids: from theory to applications



Roadmap

1. **Colloidal and non-colloidal suspensions**: size range, origin of the word "colloid"
2. **Interactions between colloids**: van der Waals attraction and irreversible aggregation
3. **Controlling aggregation**: charge and steric stabilisation, Debye–Hückel, DLVO
4. **Sheared suspensions**: Brownian vs. flow time scales, the Péclet number
5. **Origins of shear thinning**: the Krieger–Dougherty model (1959)
6. **From Krieger–Dougherty to Cross and Carreau–Yasuda**: a semi-rigorous route

Suspensions of particles

Colloidal suspensions

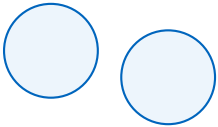
$$1\,\mathrm{nm} \leq R \leq 1\,\mu\mathrm{m}$$

Non-colloidal suspensions

$$R \geq 1\,\mu\mathrm{m}$$



colloidal



non-colloidal

Where does the word "colloid" come from?

The word *colloid* comes from the Greek word

κόλλα (glue)

and reflects the tendency of colloidal particles to aggregate irreversibly due to the attractive van der Waals interactions.

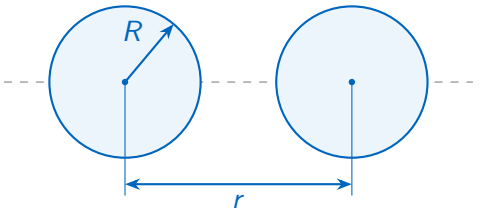
Note

Due to their intermediate (*mesoscopic*) size between the atomic scale and the macroscopic scale, colloidal particles are

- large enough to "see" the solvent molecules as a continuum,
- but small enough to experience Brownian fluctuations and to consider gravity negligible in comparison to the inter-particle and particle–solvent interactions.

(Einstein, 1905)

van der Waals potential



- r – distance between the centres of two particles
- R – radius of the colloidal particles
- A – Hamaker constant

$$U_{\mathrm{vdW}}(r) = -\frac{A}{12}\left[\frac{4R^2}{r^2-4R^2} + \frac{4R^2}{r^2} + 2\ln\left(1-\frac{4R^2}{r^2}\right)\right]$$

van der Waals potential: limiting behaviours

- For $r \gg R$ (very large distances between particles):

$$U_{\mathrm{vdW}}(r) \simeq -\frac{16}{9} A \left(\frac{R}{r}\right)^6$$

- For very close particles ($r - 2R \ll R$):

$$U_{\mathrm{vdW}}(r) \simeq -\frac{A}{12} \left(\frac{R}{r - 2R}\right)$$

If particles come very close together, $|U_{\mathrm{vdW}}| > k_B T$

$\Downarrow$

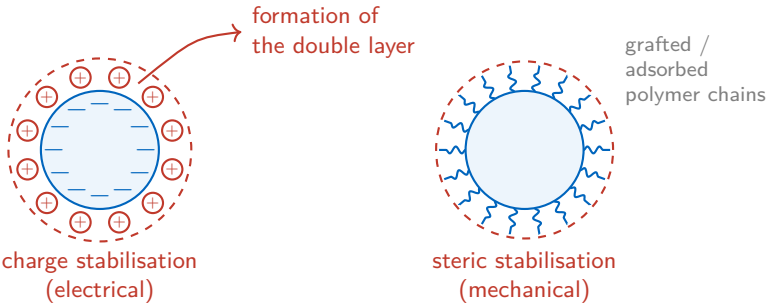
IRREVERSIBLE AGGREGATION

How to "control" particle aggregation in a colloidal system?

Equip each particle with a **repulsive shield**

- of electrostatic nature (*charge stabilisation*),
- of mechanical nature (*steric stabilisation*).

Solvent: $H_2O \rightleftharpoons H^+ + OH^-$



Charge stabilisation and the electric double layer

- Charge stabilisation is possible when colloidal particles have ionisable groups on their surface which dissociate when the particles are suspended in polar solvents.
- The charged particle is surrounded by a cloud of dissociated ions forming a double layer of ions.
- The charge distribution in the double layer results from the balance between the diffusion of the ions and the attractive Coulomb force, and generates a potential described by the Debye–Hückel equation.

Debye–Hückel potential and DLVO theory

$$V_c(r) = \frac{(Qe)^2}{4\pi\varepsilon\varepsilon_0\, r} \frac{\exp\left[-\dfrac{r-2R}{\lambda_{\mathrm{DH}}}\right]}{\left(1+\dfrac{R}{\lambda_{\mathrm{DH}}}\right)^2}$$

$$\lambda_{\mathrm{DH}} = \sqrt{\frac{\varepsilon\varepsilon_0\, k_B T}{e^2 \sum_i z_i^2\, n_i}}$$

- λ_{DH} – Debye–Hückel (screening) length
- z_i – valence of the ion species i
- n_i – corresponding number density

The full interaction potential (Derjaguin–Landau–Verwey–Overbeek, **DLVO**):

$$U_{\mathrm{DLVO}}(r) = U_{\mathrm{vdW}}(r) + V_c(r)$$

How does a uniformly sheared suspension behave?

How does a uniformly sheared suspension of colloidal (or non-colloidal) particles behave? Two time scales are involved:

(A) Diffusion time

$$\tau_B \simeq \frac{R^2}{D_0}, \qquad D_0 = \frac{k_B T}{6\pi\eta_s R}$$

D_0 – diffusion coefficient (Stokes–Einstein)

(B) Flow time scale

defined by the inverse flow rate

$$\tau_{\text{flow}} \simeq \frac{1}{\dot{\gamma}}$$

The ratio between these two time scales is the Péclet number:

$$\text{Pe} = \frac{\tau_B}{\tau_{\text{flow}}} = \tau_B\,\dot{\gamma}$$

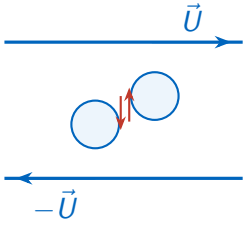
- $\text{Pe} \lesssim 1 \;\Rightarrow$ Brownian motion dominates;
- $\text{Pe} \gg 1 \;\Rightarrow$ athermal behaviour: the dynamics is determined by the flow alone.

Origins of shear-thinning behaviour in a suspension of rigid particles

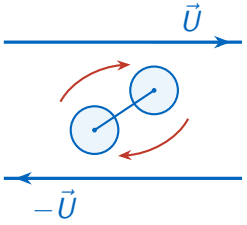
Irvin M. Krieger and Thomas J. Dougherty (1959)

Singlets and doublets in shear flow

Recall the main idea we have just discussed: it is all about a competition between shearing and thermal forces (Brownian motion).



independent rotation of particles
(**singlet** state, P_1)



rigid dumbbell motion
(**doublet** state, P_2)

In the absence of shearing forces, the distribution of singlets and doublets may be formally represented as a chemical equilibrium:

$$2P_1 \underset{k_b}{\overset{k_f}{\rightleftharpoons}} P_2$$

Shear-induced dissociation of doublets

k_f and k_b are the specific rate constants for the formation and dissociation of doublets.

Imposing a shearing force introduces a second mechanism for the decomposition of doublets, since they will be separated during the dumbbell rotation:

- k_s – specific rate constant for the shear-induced dissociation
- n_1 – number of singlets per unit volume
- n_2 – number of doublets per unit volume

$$P_2 \xrightarrow{k_s} 2P_1$$

$$\frac{\mathrm{d}n_2}{\mathrm{d}t} = k_f\, n_1^2 - (k_b + k_s)\, n_2$$

Steady-state condition:

$$\frac{\mathrm{d}n_2}{\mathrm{d}t} = 0 \implies \frac{n_2}{n_1^2} = \frac{k_f}{k_b + k_s} = \frac{k_f}{k_b}\left(1 + \frac{k_s}{k_b}\right)^{-1} \quad \circledast$$

Zero-shear equilibrium and a viscosity postulate

Because $k_s = 0$ in the absence of shear, the zero-shear concentrations of singlets and doublets, $\bar{n}_1$ and $\bar{n}_2$, satisfy

$$\frac{\bar{n}_2}{\bar{n}_1^2} = \frac{k_f}{k_b} \quad (**)$$

As a next step, we need to postulate a relationship between the viscosity of the suspension, $\eta = \tau/\dot{\gamma}$, and the concentrations of singlets and doublets:

$$\frac{\eta}{\eta_s} = f_1\, n_1 + f_2\, n_2$$

Total concentration of spheres:

$$n = n_1 + 2n_2$$

(a doublet contains two spheres)

Limiting viscosities

$$\frac{\eta}{\eta_s} = f_1\, n + (f_2 - 2f_1)\, n_2 \qquad (1)$$

Note: the factors f_1 and f_2 are assumed to be functions of n but independent of $\dot{\gamma}$.

For $\dot{\gamma} \to 0$: $n_2 \to \bar{n}_2$, $n_1 \to \bar{n}_1$, $\eta \to \eta_0$

$$\frac{\eta_0}{\eta_s} = f_1\, n + (f_2 - 2f_1)\, \bar{n}_2 \qquad (2)$$

For $\dot{\gamma} \to \infty$: all doublets will be dissociated, $n_2 \to 0$

$$\frac{\eta_\infty}{\eta_s} = f_1\, n \qquad (3)$$

$$(1) + (2) + (3) \implies \frac{\eta - \eta_\infty}{\eta_0 - \eta_\infty} = \frac{n_2}{\bar{n}_2}$$

Viscosity in terms of rate constants

From $\circledast$ and $\circledast\circledast$:

$$\frac{n_2}{\bar{n}_2} = \frac{n_1^2}{\bar{n}_1^2}\left(1 + \frac{k_s}{k_b}\right)^{-1}$$

(for a dilute population of doublets, $n_1 \simeq \bar{n}_1$)

Finally:

$$\frac{\eta - \eta_\infty}{\eta_0 - \eta_\infty} = \left(1 + \frac{k_s}{k_b}\right)^{-1}$$

$$k_s = \frac{1}{t_s}, \qquad k_b = \frac{1}{t_b}$$

- t_s – mean lifetime of a doublet dissociating only by shear forces
- t_b – mean lifetime of a doublet dissociating via thermal motion

Lifetimes of a doublet

The lifetime of a doublet dissociating under shear is the time required for the dumbbell to rotate to a position such that the two particles are in different shear layers.

Taking the angular velocity of the doublet $\Omega = \dot{\gamma}/2$ and estimating the angle of rotation as $\pi/2$:

$$t_s \simeq \left| \frac{\alpha_1 \, \pi/2}{\dot{\gamma}/2} \right| = \frac{\alpha_1 \, \pi}{|\dot{\gamma}|}$$

(α_1 reflects the approximation in the value of the angle)

The mean lifetime under thermal dissociation:

$$t_b = \frac{\overline{x^2}}{2D}$$

with

$$D = \frac{k_B T}{6\pi a \, \eta}, \qquad \eta = \frac{\tau}{\dot{\gamma}}$$

$$t_b = \frac{3\pi\eta \, a \, \overline{x^2}}{k_B T}$$

The Krieger–Dougherty result

From dimensional considerations,

$$\overline{x^2} = \alpha_2\, a^2$$

(a – radius of the particle), hence

$$t_b = \frac{3\pi\eta\,\alpha_2\, a^3}{k_B\, T}$$

Finally:

The ratio between time scales:

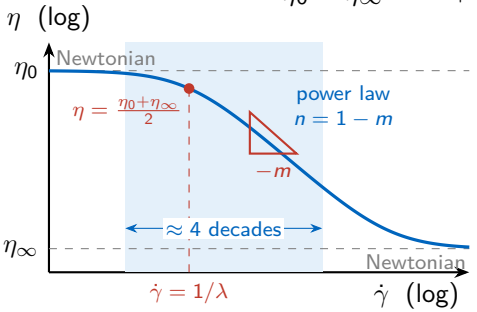
$$\frac{t_b}{t_s} = \frac{3\alpha_2\, a^3\, \eta\, |\dot{\gamma}|}{\alpha_1\, k_B\, T} = \frac{3a^3\, |\tau|}{\alpha\, k_B\, T} = \frac{|\tau|}{\tau_c}$$

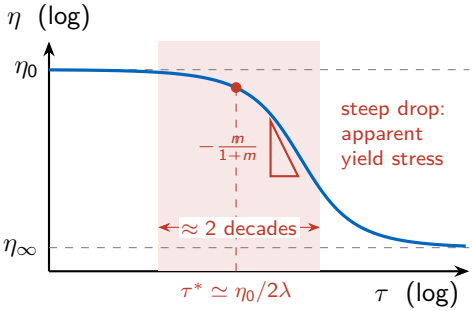
$$\alpha = \frac{\alpha_1}{\alpha_2}, \qquad \tau_c = \frac{\alpha\, k_B\, T}{3a^3}$$

$$\frac{\eta - \eta_\infty}{\eta_0 - \eta_\infty} = \left(1 + \frac{|\tau|}{\tau_c}\right)^{-1}$$

The Cross model: viscosity vs. rate of shear and vs. stress

$$\frac{\eta - \eta_\infty}{\eta_0 - \eta_\infty} = \frac{1}{1 + (\lambda\dot\gamma)^m}$$ (Cross, 1965)





Computed for $m = 0.7$, $\eta_\infty/\eta_0 = 10^{-3}$; shaded: η from $0.9\,\eta_0$ to $10^{-2}\eta_0$. As $\tau = \eta\dot\gamma \propto \dot\gamma^{\,1-m}$, the same drop is compressed in stress: $\eta \propto \dot\gamma^{-m}$ but $\eta \propto \tau^{-m/(1-m)}$.

From Krieger–Dougherty to Cross and Carreau–Yasuda

Is there a (semi-)rigorous basis for the empirical flow curves?

Step 1: from Krieger–Dougherty to Cross

The kinetic result $\dfrac{\eta-\eta_\infty}{\eta_0-\eta_\infty}=\left(1+\dfrac{k_s}{k_b}\right)^{-1}$ holds whatever the rate laws are.

Krieger–Dougherty (1959)

$k_s \propto |\dot\gamma|$; $k_b = 1/t_b$ set by the *suspension* viscosity η ($D = k_B T/6\pi a\eta$):

$$\frac{k_s}{k_b}=\frac{|\tau|}{\tau_c} \;\Rightarrow\; \text{stress-based}$$

Cross (1965)

k_b independent of the flow; shear breaks links sub-linearly, $k_s \propto |\dot\gamma|^m$, $0 < m \leq 1$:

$$\frac{k_s}{k_b}=(\lambda|\dot\gamma|)^m \;\Rightarrow\; \text{rate-based}$$

$$\frac{\eta-\eta_\infty}{\eta_0-\eta_\infty}=\frac{1}{1+(\lambda\dot\gamma)^m}$$

= Carreau–Yasuda $[1+(\lambda\dot\gamma)^a]^{(n-1)/a}$ with $a = m$, $n = 1 - m$.

Limitation: m is postulated; one exponent sets both slope and sharpness.

Step 2: a population of clusters with different lifetimes

A real suspension contains aggregates of many sizes. Let each population i carry a viscosity increment $\Delta\eta_i$, have its own thermal lifetime λ_i, and obey Krieger kinetics independently:

$$\eta(\dot\gamma) - \eta_\infty = \sum_i \frac{\Delta\eta_i}{1+\lambda_i|\dot\gamma|} \longrightarrow \int_0^\infty \frac{H(\lambda)\,\mathrm{d}\lambda}{1+\lambda|\dot\gamma|}, \qquad \int_0^\infty H\,\mathrm{d}\lambda = \eta_0 - \eta_\infty$$

Truncated power-law distribution of lifetimes ($0 < \beta < 1$, $0 < \lambda < \lambda_{\max}$):

$$H(\lambda) = (\eta_0 - \eta_\infty)\,\frac{\beta\,\lambda^{\beta-1}}{\lambda_{\max}^{\beta}}$$

With $u = \lambda|\dot\gamma|$:

$$\frac{\eta - \eta_\infty}{\eta_0 - \eta_\infty} = \frac{\beta}{(\lambda_{\max}|\dot\gamma|)^\beta} \int_0^{\lambda_{\max}|\dot\gamma|} \frac{u^{\beta-1}}{1+u}\,\mathrm{d}u$$

Step 2 (cont.): the two asymptotes

Low rates, $\lambda_{\max}|\dot{\gamma}| \ll 1$:

$$\frac{\eta - \eta_\infty}{\eta_0 - \eta_\infty} \simeq 1 - \frac{\beta}{1+\beta}\,\lambda_{\max}|\dot{\gamma}|$$

$\Rightarrow$ Newtonian plateau η_0.

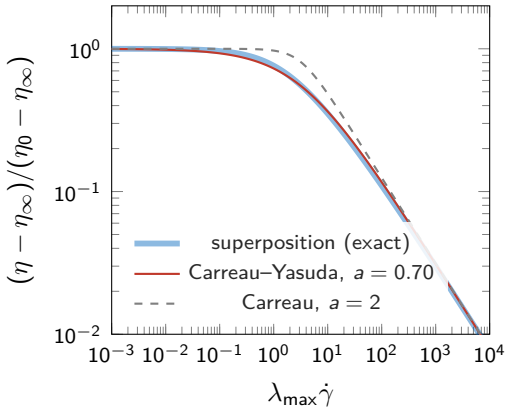
High rates, $\lambda_{\max}|\dot{\gamma}| \gg 1$:

$$\int_0^\infty \frac{u^{\beta-1}}{1+u}\,\mathrm{d}u = \frac{\pi}{\sin\pi\beta}$$

$$\frac{\eta - \eta_\infty}{\eta_0 - \eta_\infty} \simeq \frac{\beta\pi}{\sin\pi\beta}\,(\lambda_{\max}|\dot{\gamma}|)^{-\beta}$$

Power law with $n = 1 - \beta$: no longer postulated but **derived**.
It is set by the breadth of the distribution of cluster lifetimes.

Step 3: Carreau–Yasuda as a closed-form approximation



The superposition can be done exactly:

$$\frac{\eta-\eta_\infty}{\eta_0-\eta_\infty} = {}_2F_1(1,\beta;1+\beta;-\lambda_{\max}|\dot{\gamma}|)$$

Carreau–Yasuda with

- $n = 1-\beta$ (high-rate asymptote),
- $\lambda = \lambda_{\max}\,(\beta\pi/\sin\pi\beta)^{-1/\beta}$ (prefactor),
- a fitted to the crossover,

reproduces it within 6 % for $\beta = 0.6$ ($a \simeq 0.70$, $\lambda \simeq 0.32\,\lambda_{\max}$). Carreau ($a = 2$) is off by up to 46 %.

What is rigorous, and what is not

η_0, η_∞	all links intact / all links broken
λ	longest (thermal) cluster lifetime
n	$1-\beta$: breadth of the lifetime distribution
a	shape of the crossover, i.e. how the distribution is cut off near $\lambda_{\max}$: phenomenological (Yasuda *et al.*, 1981)

The kinetic kernel $1/(1+\lambda|\dot\gamma|)$ depends on $|\dot\gamma|$, so it is not analytic at rest: this is why breakdown models of suspensions give $a \lesssim 1$ rather than $a = 2$.

Why $a = 2$ for polymer melts and solutions

- Frame invariance: η depends on $\mathrm{II_D} \propto \dot\gamma^2$. A fluid that is analytic near rest has $\eta = \eta_0 - c\,\dot\gamma^2 + O(\dot\gamma^4)$, hence $a = 2$.
- One Maxwell mode + Cox–Merz: $|\eta^*| = \eta_0/\sqrt{1+(\lambda\omega)^2}$, exactly Carreau with $n = 0$, $a = 2$.
- Carreau (1972) derived it from a network theory.

Krieger & Dougherty, *Trans. Soc. Rheol.* 3, 137 (1959); Cross, *J. Colloid Sci.* 20, 417 (1965); Carreau, *Trans. Soc. Rheol.* 16, 99 (1972); Yasuda, Armstrong & Cohen, *Rheol. Acta* 20, 163 (1981); Cox & Merz, *J. Polym. Sci.* 28, 619 (1958).